A Synthetic Multi-Store Retail Point-of-Sale Transaction Dataset in Square POS Export Format

Dataset: A Synthetic Multi-Store Retail Point-of-Sale Transaction Dataset in Square POS Export Format (30 Stores, 24 Months)

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Author contributions: Conceptualization, dataset design, data generation, validation, and documentation.
Version: 2.0 | **Date:** July 11, 2026
Coverage: January 1, 2024 – December 31, 2025 (731 calendar days, 729 selling days)
Scale: 970,838 sales line items | 30 store locations | 32 items / 67 SKUs / 6 categories | ~$49.3M gross sales

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Part A — Dataset Documentation

A1. Introduction

This dataset is a fully synthetic collection of retail point-of-sale (POS) transaction records modeled on the export formats of Square POS, a widely used payment and retail management platform. It simulates a 30-store US retail chain over two complete calendar years, consolidated into a single line-item-level report in the structure of Square's "Items Detail CSV" export.

The dataset was procedurally generated by a seeded simulation script and contains no real business, customer, or personal data. Its distinguishing feature is a documented **ground-truth retail event calendar**: every demand influence embedded in the data (holidays, Black Friday, Cyber Monday, Super Saturday, seasonal ramps, store closures, post-holiday returns) is recorded in an accompanying file, enabling users to verify whether analytical or machine-learning methods correctly recover known effects. This makes the dataset suitable as a benchmark for retail analytics, time-series forecasting, interpretability validation, and teaching.

A2. Dataset Files

File	Content	Rows	Size
square_item_sales_detail_24mo.csv (.gz)	Main dataset: transaction line items	970,838	150 MB (18 MB gzipped)
ground_truth_event_calendar.csv	Per-day applied multipliers and events	731	38 KB
square_item_sales_summary_by_store_24mo.csv	Item x variation x store aggregation	2,010	219 KB
square_sales_summary_by_store_24mo.csv	One summary row per store	30	2.5 KB
validation_report.json	Realized dataset statistics	-	1 KB
generate_square_dataset_v2.py	Seeded generation script	-	14 KB

The gzipped detail file can be read directly, e.g. `pandas.read_csv("square_item_sales_detail_24mo.csv.gz")`.

A3. Dataset Characteristics

Property	**Value**
Reporting period	2024-01-01 to 2025-12-31 (two complete calendar years; 2024 is a leap year)
Selling days	729 of 731 (stores closed on Christmas Day in both years)
Stores (Locations)	30, named “Store 01 - Austin” … “Store 30 - Kansas City”, in three volume tiers
Product catalog	32 items, 67 SKUs (size/color/capacity variations), 6 categories
Categories	Apparel, Footwear, Accessories, Electronics, Home Goods, Beauty
Event types	Payment: 961,389 rows; Refund: 9,449 rows
Currency / tax	USD; flat 8.25% sales tax on net sales
Total gross sales	$49.27 million across the chain
Year-over-year growth	2025 net sales are 5.5% above 2024 (realized)
Format reference	Square “Items Detail CSV” export structure (one row per item line, repeated Transaction ID)

A4. Data Dictionary (Detail File)

One row represents one item line within a transaction; rows sharing a Transaction ID form one basket.

#	Column	Type	Description
1	Date	Date	Transaction date
2	Time	Time (HH:MM)	Transaction time, 24-hour clock
3	Time Zone	Text	Store time zone (CST, simplified)
4	Category	Text	Item category (one of 6)
5	Item	Text	Item name
6	Qty	Integer	Units in the line; negative for refunds
7	Price Point Name	Text	Variation (size, color, capacity)
8	SKU	Text	Stock keeping unit identifier
9	Unit Price	Decimal	Price per unit at time of sale
10	Gross Sales	Decimal	Unit Price x Qty, before discounts
11	Discounts	Decimal	Discount amount, stored as a negative value
12	Net Sales	Decimal	Gross Sales + Discounts
13	Tax	Decimal	8.25% of Net Sales
14	Total Collected	Decimal	Net Sales + Tax
15	Transaction ID	Text	Basket identifier (e.g., TXN1000001)
16	Payment Method	Text	Card, Cash, or Other
17	Device Name	Text	POS device (Register 1, Register 2, iPad POS)
18	Location	Text	Store name; the multi-store key
19	Customer ID	Text	Present for ~35% of transactions (identified customers)
20	Event Type	Text	Payment or Refund
21	Unit Cost	Decimal	Cost of goods per unit
22	Gross Profit	Decimal	Net Sales - (Unit Cost x Qty)

Row-level accounting identities hold throughout: Net Sales = Gross Sales + Discounts, and Total Collected = Net Sales + Tax. Refund rows carry negative Qty and negative monetary values, consistent with Square's representation. Columns 1-8, 10-15, 17-18, and 20 correspond to fields in Square's real exports; Unit Cost and Gross Profit mirror the cost-of-goods reporting available in Square for Retail.

A5. Embedded Retail Event Calendar

The dataset's demand is shaped by a two-year US retail calendar. Every effect below is applied programmatically and recorded per day in ground_truth_event_calendar.csv.

Event	Dates (2024 / 2025)	Traffic effect	Additional effects
New Year's Day	Jan 1	0.5x	-
January slump	All January	0.8x month factor	Elevated returns Jan 2-6
Valentine's Day (+eve)	Feb 13-14	up to 1.55x	Beauty 2.0x, Accessories 1.7x demand
Easter	Mar 31 / Apr 20	0.3x (mostly closed)	-
Mother's Day (+eve)	May 11-12 / May 10-11	up to 1.40x	Beauty, Home Goods, Accessories skew
Memorial Day	May 27 / May 26	1.35x	Promotional discounting (45% of lines, up to 25%)
Father's Day (+eve)	Jun 15-16 / Jun 14-15	up to 1.30x	Electronics 1.7x skew
Independence Day	Jul 4	1.35x	Promotional discounting
Back-to-school	All August	1.12x month factor	Apparel 1.4x, Footwear 1.35x skew
Labor Day	Sep 2 / Sep 1	1.35x	Promotional discounting
Thanksgiving	Nov 28 / Nov 27	0.15x (mostly closed)	-
Black Friday	Nov 29 / Nov 28	**2.8x**	Electronics 2.0x skew; 65% of lines discounted, depths to 40%
BF Weekend	Sat-Sun after	1.7x / 1.5x	Continued Electronics skew and discounts
Cyber Monday	Dec 2 / Dec 1	1.35x	Electronics 2.2x skew; deep discounts. Modest in-store traffic reflects its online-centric nature
December ramp	Dec 1-23	month factor rising to ~1.6x	Season-wide discounting (40% of lines)
Super Saturday	Dec 21 / Dec 20	**2.3x**	Combined with the December ramp, the highest-revenue days of each year
Christmas Eve	Dec 24	1.6x	-
Christmas Day	Dec 25	**Closed (zero sales)**	-
Post-Christmas returns	Dec 26-31	1.35x	Refund probability raised ~3.5x; deep clearance discounts
New Year's Eve	Dec 31	1.2x	-

Baseline (non-event) behavior includes: day-of-week multipliers of 0.90 / 0.85 / 0.90 / 1.00 / 1.25 / 1.55 / 1.35 (Monday-Sunday); a mild annual sinusoidal cycle (+/-8%); month-level adjustments; and a +6% year-over-year growth factor applied to 2025.

A note on realism: in the generated data, Super Saturday exceeds Black Friday as the top revenue day of each year. This is a deliberate design choice reflecting brick-and-mortar POS behavior, where Black Friday spending has increasingly shifted to online channels while Super Saturday remains an in-store peak; the December ramp compounds the Super Saturday multiplier.

A6. Realized Statistics (Validation Summary)

Because random sampling means realized values can differ from configured parameters, the following statistics were measured from the generated data and should be treated as the authoritative description:

Statistic	Realized value
Total rows	970,838 (961,389 payments; 9,449 refunds)
Selling days	729 (zero rows on 2024-12-25 and 2025-12-25)
Discounted line share	32.5% of payment lines
Card payment share	71.0% of transactions
Customer ID attribution	35.0% of transactions
Year-over-year net sales growth	+5.5% (2025 vs 2024; the 2024 leap day slightly dampens the configured +6%)
Top 5 revenue days	Super Saturday 2025 (~$298K), Super Saturday 2024 (~$276K), Black Friday 2025 (~$187K), Black Friday 2024 (~$170K), Dec 27 2025 (~$167K)

A7. Suggested Uses and Benchmark Tasks

1. **Time-series forecasting** with proper chronological evaluation, e.g., train on 2024, test on 2025; or train through September 2025 and test on the holiday quarter. Two full years provide two instances of every calendar event.
2. **Event-effect estimation**: estimate holiday uplifts, then verify estimates against the ground-truth calendar.
3. **Interpretability validation**: test whether feature-attribution methods (e.g., SHAP) recover the known drivers (day of week, event days, category skews) encoded in generation.
4. **Store clustering and performance analysis** across the three designed volume tiers.
5. **Market-basket analysis** via Transaction ID groupings.
6. **Margin and discount analytics** using Unit Cost, Gross Profit, and event-driven discount depths.
7. **Anomaly detection**: closures, spikes, and returns-week refund surges are labeled ground truth.
8. **Teaching**: a realistic, privacy-free substitute for commercial POS exports.

A8. Provenance, Disclosure, and Disclaimer

This dataset is fully synthetic. All values were produced by the accompanying seeded Python simulation script (procedural generation using NumPy pseudo-random sampling); no values were authored by, or copied from, any real business system. The generation script and documentation were developed with the assistance of an AI system, with all design parameters, validation, and final decisions reviewed by the dataset author, MD Owes Quruny Shubho (ORCID: 0009-0002-4870-5280). Users should consult the policies of individual repositories and journals regarding AI-assisted research artifacts.

Square is a product of Block, Inc. This dataset replicates only the publicly documented structure of Square's export files for interoperability and educational purposes and is not affiliated with, endorsed by, or derived from Square or Block, Inc. systems or data.

A9. Known Limitations

1. Simplified column set relative to Square's full 52-column Transactions CSV: processing fees, payout identifiers, card brand, PAN suffix, and entry-method fields are not modeled; bank-settlement analysis is out of scope.
2. A single flat tax rate (8.25%) and a single time zone label (CST) are applied to all stores, whereas a real 30-store US chain would span multiple jurisdictions.
3. Customer IDs identify repeat customers only nominally; no demographic, loyalty-tier, or lifetime-value structure is modeled, and ~65% of transactions are anonymous.
4. Prices are constant across the two years (no inflation or price-change events); demand growth is modeled through traffic, not pricing.
5. Item catalog is static: no new-product introductions or discontinuations occur during the period.
6. Weather, local events, and store-specific idiosyncrasies (renovations, openings) are not modeled.
7. Online/omnichannel sales are excluded by design; the data represents in-store POS activity only, which is why Cyber Monday shows only a modest in-store effect.

Part B — Data Generation Methodology

B1. Tools and Reproducibility

The dataset is produced by a single Python script, `generate_square_dataset_v2.py`, using pandas and NumPy. All randomness flows from one seeded generator (`numpy.random.default_rng(42)`), making every run bit-for-bit reproducible. Running the script regenerates the identical dataset, all aggregate files, the ground-truth calendar, and the validation report.

B2. Pipeline Overview

The generation pipeline has six stages:

1. **Entity setup** — stores with volume tiers; the item catalog with prices, costs, and popularity weights.
2. **Calendar construction** — a programmatic two-year US retail event calendar defining, for every date, the traffic multiplier, category demand skews, discount profile, refund multiplier, and closure flag.
3. **Demand simulation** — Poisson-distributed transaction counts per store-day.
4. **Transaction composition** — basket construction, timing, payment method, device, and customer attribution.
5. **Financial computation** — line-level gross/discount/net/tax/profit values and refund injection.
6. **Aggregation and validation** — derived summary files, the ground-truth calendar export, and a realized-statistics report.

B3. Stage 1 — Entity Setup

Stores. Thirty locations are defined across US cities. Each store is randomly assigned a volume tier: large (multiplier 1.8, probability 0.2), medium (1.0, probability 0.5), or small (0.55, probability 0.3). Tiers are fixed for the full two years, creating a stable, discoverable store-size structure.

Catalog. Thirty-two items across six categories carry one or more variations (sizes, colors, capacities), yielding 67 SKUs. Each SKU has a unit price, a unit cost of 28-55% of price (category-dependent, so gross margins differ systematically by category), and a popularity weight from 0.6 (Leather Boots) to 2.5 (Cotton Crew T-Shirt), producing a realistic long-tail sales distribution.

B4. Stage 2 — Calendar Construction

Floating US holidays are computed programmatically: Thanksgiving as the fourth Thursday of November (Black Friday the day after; Cyber Monday the Monday after), Mother’s Day as the second Sunday of May, Father’s Day as the third Sunday of June, Memorial Day as the last Monday of May, Labor Day as the first Monday of September, and Super Saturday as the last Saturday before December 25. Easter dates are supplied per year. Each calendar entry specifies up to five effects: a traffic multiplier, category demand skews, a discount profile (probability and depth set), a refund-probability multiplier, and a closure flag. When events collide on one date, the strongest traffic effect prevails. The full realized calendar is exported as ground_truth_event_calendar.csv.

B5. Stage 3 — Demand Simulation

For each store s and date d, the transaction count is drawn as:

$$N(s, d) \sim \text{Poisson}(22 \times \text{tier}(s) \times \text{dow}(d) \times \text{season}(d) \times \text{month}(d) \times \text{event}(d) \times \text{yoy}(d))$$

where dow is the day-of-week multiplier (Monday-Sunday: 0.90, 0.85, 0.90, 1.00, 1.25, 1.55, 1.35), season(d) = 1 + 0.08 sin(2 pi yday / 365.25), month(d) is the month-level factor (January 0.80 through a December ramp of 1.15 rising to ~1.60 by December 23), event(d) is the calendar multiplier from Stage 2, and yoy(d) = 1.06 for dates in 2025. Closed days (Christmas) generate zero transactions.

B6. Stage 4 — Transaction Composition

Each transaction contains 1-4 line items (probabilities 0.55 / 0.28 / 0.12 / 0.05) sampled without replacement from the catalog. Sampling weights are the base popularity weights multiplied by the day's category skews (e.g., Electronics x2.0 on Black Friday; Apparel x1.4 in August) and renormalized — so events change both how much sells and what sells. Timestamps are drawn from a normal distribution centered at 14:30, clipped to store hours (09:00-20:00). Payment method is sampled as Card 71% / Cash 24% / Other 5%; one of three POS devices is assigned; a Customer ID is attached with probability 0.35 from a pool of 20,000 customer identifiers.

B7. Stage 5 — Financial Computation and Refunds

Line quantities are sampled from {1, 2, 3} with weights favoring single units (80% singles, 17% pairs, 3% triples). Then: Gross Sales = Unit Price x Qty; a discount is applied with the day's probability (baseline 30%; December 40%; promotional events 45-65%) at the day's depth set (baseline 10-20%; Black Friday and Cyber Monday up to 40%), stored as a negative Discounts value; Net Sales = Gross Sales + Discounts; Tax = 8.25% of Net Sales; Total Collected = Net Sales + Tax; Gross Profit = Net Sales - Unit Cost x Qty.

After each transaction, a refund event is injected with baseline probability 1.5%, scaled by the day's refund multiplier (~3.5x during December 26-31 and ~2.5x in early January, capped at 6%). Refunds are separate rows with their own Transaction ID, Event Type "Refund", quantity -1, and negated monetary values.

B8. Stage 6 — Aggregation and Validation

Two summary files are derived with pandas groupby operations: item sales by Location x Category x Item x SKU x Variation, and a per-store sales summary. A validation report is then computed from the generated data and saved as validation_report.json, covering row counts by event type, date coverage, realized discount/card/customer-attribution rates, total gross sales, the top five revenue days, Christmas-closure verification, and realized year-over-year growth. The realized values (Section A6) are reported in documentation in preference to configured parameters, eliminating any gap between documented and actual generation behavior.

B9. Design Rationale

1. **Two complete calendar years** provide every calendar event twice, enabling train-on-year-1 / test-on-year-2 evaluation designs and year-over-year comparisons that a single quarter cannot support.
2. **Past dates (2024-2025)** avoid the oddity of a public dataset containing future-dated sales.
3. **A ground-truth calendar** converts the dataset from sample data into a benchmark: methods can be scored on whether they recover known effects.
4. **Event-conditional discounting and refunds** reproduce the joint movement of traffic, promotion depth, and returns that characterizes real retail — patterns that simple seasonal noise cannot produce.
5. **In-store framing** motivates the deliberate Cyber Monday attenuation and Super Saturday prominence.

B10. Customization

All behavior is controlled by configurable parameters, including the store list, item catalog, date range, daily transaction rate, seasonal and event multipliers, discount and refund settings, customer attribution rate, tax rate, and random seed. Changing the seed generates a statistically similar but unique dataset, while expanding the store list or item catalog naturally scales the dataset.

B11. References

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7. Square Developer — Sample sales report application: developer.squareup.com/docs/payments/scenarios/simple-sales-report

B12. Contact

Questions about this dataset, its generation methodology, or reuse may be directed to the author:

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